

Total No. of Questions - [3]

Total No. of Printed Pages: 02

G.R. No.

PAPER CODE

223-253 (PSE)

May 2023 (ENDSEM) EXAM

S.Y. (IT) (AY 2022-23 SEMESTER - II)

COURSE NAME: COMPUTER NETWORKS

COURSE CODE: ITUA22203

(PATTERN 2020)

Time: [1Hr]

[Max. Marks: 30]

Instructions to candidates:

- 1) Use of scientific calculator is allowed
- 2) Use suitable data where ever required
- 3) All questions are compulsory

| Q. No. | Question Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Max. Marks | CO mapped | BT Level |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|----------|
| Q.1    | <p>a) Design and explain the algorithm for creating a numbered Access List to filter on source address</p> <p>b) Design the algorithm for updating routing table on receipt of RIP response message from neighbor router. The router has its old routing table as shown in fig 1. It receives RIP message from router C, estimate the new routing table</p> <div style="text-align: center;"> <pre> graph LR     subgraph "RIP message from C"         direction TB         R1[Net2: 4]         R2[Net3: 8]         R3[Net6: 4]         R4[Net8: 3]         R5[Net9: 5]     end     subgraph "RIP message from C after increment"         direction TB         R6[Net2: 5]         R7[Net3: 9]         R8[Net6: 5]         R9[Net8: 4]         R10[Net9: 6]     end     subgraph "Old routing table"         direction TB         O1[Net1: 7, A]         O2[Net2: 2, C]         O3[Net6: 8, F]         O4[Net8: 4, E]         O5[Net9: 4, F]     end     subgraph "New routing table"         direction TB         N1[Net   No. of hops   Next Hop]     end     O1 --&gt; UA[Updating algorithm]     O2 --&gt; UA     O3 --&gt; UA     O4 --&gt; UA     O5 --&gt; UA     R1 --&gt; UA     R2 --&gt; UA     R3 --&gt; UA     R4 --&gt; UA     R5 --&gt; UA     UA --&gt; N1         </pre> <p><b>Fig.1 Example of Updating a Routing Table</b></p> </div> | [4]        | [ 4 ]     | [ 4 ]    |
|        | <b>OR</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |            |           |          |
|        | c) Explain any TWO type of links in OSPF routing protocol, also compare both on different parameters                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | [6]        | [ 4 ]     | [ 4 ]    |

|     |                                                                                                                                      |     |       |       |
|-----|--------------------------------------------------------------------------------------------------------------------------------------|-----|-------|-------|
| Q.2 | a) What are the uses of setting Access Control Lists (ACL) rules at routers? Also, Comment on direction and location to apply ACL's. | [4] | [ 5 ] | [ 2 ] |
|     | b) Demonstrate the process of configuring SSH, interpret each parameter used in every command.                                       | [6] | [ 5 ] | [ 3 ] |
|     | <b>OR</b>                                                                                                                            |     |       |       |
| Q.3 | c) What are the benefits of VLAN? Explain any four among it.                                                                         | [6] | [ 5 ] | [ 3 ] |
|     | a) Explain various ISDN devices and interfaces using figure                                                                          | [4] | [ 6 ] | [ 3 ] |
|     | b) Draw LAPD frame structure of ISDN network, and explain address field in detail                                                    | [6] | [ 6 ] | [ 3 ] |
|     | <b>OR</b>                                                                                                                            |     |       |       |
|     | c) Consider ATM adaptation layer receives constant bit rate data from the upper layer, construct SAR PDU using AAL1                  | [6] | [ 6 ] | [ 3 ] |

**Note:** [BT Level- 1.Remember 2.Understand 3.Apply 4.Analyze 5.Evaluate 6.Create]